



# 9425 - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

Cycle: 4, Proposal Category: GO

## INVESTIGATORS

| <i>Name</i>                             | <i>Institution</i>         |
|-----------------------------------------|----------------------------|
| <b>Sander Schouws (PI) (ESA Member)</b> | <b>Universiteit Leiden</b> |

## OBSERVATIONS

| <i>Folder</i>          | <i>Observation</i> | <i>Label</i>  | <i>Observing Template</i>        | <i>Science Target</i> |
|------------------------|--------------------|---------------|----------------------------------|-----------------------|
| MIRI LRS CAPERS-126973 |                    |               |                                  |                       |
|                        | 1                  | CAPERS-126973 | MIRI Low Resolution Spectroscopy | (1) CAPERS-126973     |
|                        | 2                  | CAPERS-126973 | MIRI Low Resolution Spectroscopy | (1) CAPERS-126973     |
| MIRI LRS Panoramic-z15 |                    |               |                                  |                       |
|                        | 3                  | Panoramic-z15 | MIRI Low Resolution Spectroscopy | (3) Panoramic-z15     |
|                        | 4                  | Panoramic-z15 | MIRI Low Resolution Spectroscopy | (3) Panoramic-z15     |

## ABSTRACT

ALMA proposal ID: 2025.1.01606.L

One of JWST's most exciting discoveries is the unexpectedly large population of bright galaxies at cosmic dawn, suggesting a rapid buildup that challenges current formation models.

However, this emerging picture is largely based on rest-UV/optical light, lacking critical insights into obscured star formation, dust content and ISM conditions.

To address this crucial gap, we propose PHOENIX, an efficient and transformative ALMA Large Program targeting the diagnostic [OIII]88um line

and dust continuum emission in a carefully

selected, representative sample of 15 intrinsically luminous galaxies at  $8 < z < 15$ , quadrupling current samples.

By combining ALMA's unique capabilities with rich ancillary JWST data, PHOENIX will:

- 1) Reveal the fraction of dust-obscured star formation (SF), providing the first robust census of total (UV+IR) SF at  $z > 8$
- 2) Unveil the emergence of dust and distinguish between dominant dust production mechanisms (SN vs. grain growth)
- 3) Characterize the ISM conditions fueling early galaxy growth

PHOENIX will produce a landmark dataset, unlocking key insights into galaxy formation during cosmic dawn and illuminating this pivotal yet still poorly understood epoch.

## **OBSERVING DESCRIPTION**

(Copied from PDF attachment). We target 2 very high redshift galaxies with MIRI LRS, covering rest-frame optical emission lines in particular [OIII]5007,4959, Hbeta and Halpha. Our targets are Panoramic-z15 at  $z=14.5$  and CAPERS-126973 at  $z=11.01$ .

We estimate expected fluxes for each of our targets both based on SED fitting + ISM modeling as well as previous detections of high redshift galaxies with MIRI LRS. We set the requested sensitivity to ensure  $S/N > 5\sigma$  for the most prominent emission lines in the spectrum ([OIII]5007 and Halpha) with conservative assumptions. For the sensitivity calculations we use a sersic profile with an extent of  $0.1''$ , consistent with the source-sizes derived from the NIRCам imaging available for our sources. For a check of our strategy, we perform the same estimates for similar targets as our galaxies in the upcoming programmes GO-8051 and 9165 and verify that our strategy results in similar total integration times as requested by those programmes.

We use the FASTR1 readout pattern with 149 groups with either 9 or 12 integrations/exp for Panoramic-z15 and CAPERS-126973 respectively. This ensures that each integration is below 500 seconds to mitigate the effects from cosmic ray showers. The total integration times are 29,776s for Panoramic-z15 and 39,696s for CAPERS-126973.

For the dithering we follow GO-8051, employing a 2 dither 'ALONG SLIT NOD' and a 2-column mosaic with large overlap. This effectively results in 4 dither positions for better mitigation of bad/hot pixels and improved reconstruction of backgrounds.

## JWST Proposal 9425 (Created: Monday, November 10, 2025, 6:00:30PM Eastern Standard Time) - Overview

To ensure sub-pixel accuracy for the placement of our targets in the slit, we employ an offset TA strategy using close-by K stars from GAIA-DR3. We find suitable TA stars for each of our targets:

Panoramic-z15; Gaia DR3 2679153595122839296; Separation from main science target: 31.6 arcsec; Mag\_G = 13.4 mag; Teff = 5300 K, log g = 4.1

CAPERS-126973; Gaia DR3 2489665895791256320; Separation from main science target: 39.7 arcsec; Mag\_G = 17.6 mag; Teff = 4700 K, log g = 4.6

Using the ETC we verify that both TA stars can be detected with short integration times at  $S/N \gg 20$  without saturating the detector.

In general, the observing strategy (i.e. total integration times, exposure parameters, dithering, TA etc.) employed in this programme are very similar to successful programme GO-3703 (which detected [OIII] and H $\alpha$  in GHZ2 at  $z=12.33$ ) and accepted Cycle-4 GO programmes that target similar  $z > 9.5$  galaxies (GO-8051, 8544 and 9165).

Our targets are very compact ( $R_{\text{eff}} \sim 0.1''$ ) and only occupy a small fraction of the slit. Furthermore our employed dithering strategy with 'ALONG SLIT NOD' as well as a 2-column mosaic results in 4 dither positions rather than 2, allowing for a better characterization of the backgrounds. No dedicated background observations are necessary.

Our requested observations are sensitive to background levels, impacting the expected S/N by more than the 5% threshold described in the JWST user documentation (<https://jwst-docs.stsci.edu/jwst-general-support/jwst-background-model/jwst-background-limited-observations#gsc.tab=0>). Therefore we set a special requirements for background limited observations (consistent with accepted programmes GO 3703 and GO 8051). We request the lowest background percentile that can still be scheduled without requiring observations in the micro-meteoroid avoidance zone (10% and 30% for CAPERS-126973 and Panoramic-z15 respectively).

# Proposal 9425 - Targets - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

| Fixed Targets                                                                                                                                                                          | #                                                                                                                                                                                      | Name             | Target Coordinates                                                                | Targ. Coord. Corrections                                                                                             | Miscellaneous |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------|
|                                                                                                                                                                                        | (1)                                                                                                                                                                                    | CAPERS-126973    | RA: 02 17 3.4622 (34.2644258d)<br>Dec: -05 05 46.55 (-5.09626d)<br>Equinox: J2000 |                                                                                                                      |               |
|                                                                                                                                                                                        | Comments:<br>Category=Galaxy<br>Description=[High-redshift galaxies]                                                                                                                   |                  |                                                                                   |                                                                                                                      |               |
|                                                                                                                                                                                        | (2)                                                                                                                                                                                    | TA_CAPERS-126973 | RA: 02 17 6.0867 (34.2753612d)<br>Dec: -05 05 53.01 (-5.09806d)<br>Equinox: J2000 | Proper Motion RA: -4.078 mas/yr<br>Proper Motion Dec: -0.409 mas/yr<br>Parallax: 0.00089"<br>Epoch of Position: 2016 |               |
|                                                                                                                                                                                        | Comments: Gaia DR3 2489665895791256320<br>Seperation from main science target: 39.7 arcsec<br>Mag_G = 17.6 mag<br>Teff = 4700 K, log g = 4.6<br>Category=Star<br>Description=[K stars] |                  |                                                                                   |                                                                                                                      |               |
|                                                                                                                                                                                        | (3)                                                                                                                                                                                    | Panoramic-z15    | RA: 22 17 0.0874 (334.2503642d)<br>Dec: +00 22 45.13 (.37920d)<br>Equinox: J2000  |                                                                                                                      |               |
|                                                                                                                                                                                        | Comments:<br>Category=Galaxy<br>Description=[High-redshift galaxies]                                                                                                                   |                  |                                                                                   |                                                                                                                      |               |
|                                                                                                                                                                                        | (4)                                                                                                                                                                                    | TA_Panoramic-z15 | RA: 22 16 58.0092 (334.2417050d)<br>Dec: +00 22 44.93 (.37915d)<br>Equinox: J2000 | Proper Motion RA: 16.605 mas/yr<br>Proper Motion Dec: -0.921 mas/yr<br>Parallax: 0.00119"<br>Epoch of Position: 2016 |               |
| Comments: Gaia DR3 2679153595122839296<br>Seperation from main science target: 31.6 arcsec<br>Mag_G = 13.4 mag<br>Teff = 5300 K, log g = 4.1<br>Category=Star<br>Description=[K stars] |                                                                                                                                                                                        |                  |                                                                                   |                                                                                                                      |               |

# Proposal 9425 - Observation 1 - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

|                   |                                                                                             |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
|-------------------|---------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------|--------------------------|----------------------------|---------------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 9425, Observation 1: CAPERS-126973                                                 |                    |                                                                                   |                          |                            |                     |                    |                     | Mon Nov 10 23:00:30 GMT 2025 |
|                   | Diagnostic Status: Warning                                                                  |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
|                   | Observing Template: MIRI Low Resolution Spectroscopy                                        |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
| Diagnostics       | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
| Fixed Targets     | #                                                                                           | Name               | Target Coordinates                                                                | Targ. Coord. Corrections |                            |                     | Miscellaneous      |                     |                              |
|                   | (1)                                                                                         | CAPERS-126973      | RA: 02 17 3.4622 (34.2644258d)<br>Dec: -05 05 46.55 (-5.09626d)<br>Equinox: J2000 |                          |                            |                     |                    |                     |                              |
|                   | Comments:<br>Category=Galaxy<br>Description=[High-redshift galaxies]                        |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
|                   |                                                                                             |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
| Acquisition       | #                                                                                           | Target             | Filter                                                                            | Readout Pattern          | Groups/Int                 | Integrations/Exp    | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                           | 2 TA_CAPERS-126973 | F560W                                                                             | FAST                     | 8                          | 1                   | 1                  | 22.2                | 251072                       |
| Template          | Subarray                                                                                    |                    |                                                                                   |                          | Obtain Verification Image? |                     |                    |                     |                              |
|                   | FULL                                                                                        |                    |                                                                                   |                          | false                      |                     |                    |                     |                              |
| Mosaic            | Rows                                                                                        | Columns            | Row Overlap %                                                                     | Column Overlap %         | Row shift (deg)            | Column shift (deg)  | Tile Order         |                     |                              |
|                   | 1                                                                                           | 2                  | 0.0                                                                               | 80.0                     | 0.0                        | 0.0                 | DEFAULT            |                     |                              |
| Dithers           | #                                                                                           | Dither Type        | No. Spectral Steps                                                                | Spectral Step Offset     | No. Spatial Steps          | Spatial Step Offset |                    |                     |                              |
|                   | 1                                                                                           | ALONG SLIT NOD     |                                                                                   |                          |                            |                     |                    |                     |                              |
| Spectral Elements | #                                                                                           | Readout Pattern    | Groups/Int                                                                        | Integrations/Exp         | Total Integrations         | Exposures/Dith      | Total Dithers      | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                           | FASTR1             | 111                                                                               | 16                       | 32                         | 1                   | 2                  | 9940.193            |                              |

|                      |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| Special Requirements | Background Limited. Background no more than 10th percentile above minimum |
|----------------------|---------------------------------------------------------------------------|

# Proposal 9425 - Observation 2 - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

|                   |                                                                                             |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
|-------------------|---------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------|--------------------------|----------------------------|---------------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 9425, Observation 2: CAPERS-126973                                                 |                    |                                                                                   |                          |                            |                     |                    |                     | Mon Nov 10 23:00:30 GMT 2025 |
|                   | Diagnostic Status: Warning                                                                  |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
|                   | Observing Template: MIRI Low Resolution Spectroscopy                                        |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
| Diagnostics       | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
| Fixed Targets     | #                                                                                           | Name               | Target Coordinates                                                                | Targ. Coord. Corrections |                            |                     | Miscellaneous      |                     |                              |
|                   | (1)                                                                                         | CAPERS-126973      | RA: 02 17 3.4622 (34.2644258d)<br>Dec: -05 05 46.55 (-5.09626d)<br>Equinox: J2000 |                          |                            |                     |                    |                     |                              |
|                   | Comments:<br>Category=Galaxy<br>Description=[High-redshift galaxies]                        |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
|                   |                                                                                             |                    |                                                                                   |                          |                            |                     |                    |                     |                              |
| Acquisition       | #                                                                                           | Target             | Filter                                                                            | Readout Pattern          | Groups/Int                 | Integrations/Exp    | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                           | 2 TA_CAPERS-126973 | F560W                                                                             | FAST                     | 8                          | 1                   | 1                  | 22.2                | 251072                       |
| Template          | Subarray                                                                                    |                    |                                                                                   |                          | Obtain Verification Image? |                     |                    |                     |                              |
|                   | FULL                                                                                        |                    |                                                                                   |                          | false                      |                     |                    |                     |                              |
| Mosaic            | Rows                                                                                        | Columns            | Row Overlap %                                                                     | Column Overlap %         | Row shift (deg)            | Column shift (deg)  | Tile Order         |                     |                              |
|                   | 1                                                                                           | 2                  | 0.0                                                                               | 80.0                     | 0.0                        | 0.0                 | DEFAULT            |                     |                              |
| Dithers           | #                                                                                           | Dither Type        | No. Spectral Steps                                                                | Spectral Step Offset     | No. Spatial Steps          | Spatial Step Offset |                    |                     |                              |
|                   | 1                                                                                           | ALONG SLIT NOD     |                                                                                   |                          |                            |                     |                    |                     |                              |
| Spectral Elements | #                                                                                           | Readout Pattern    | Groups/Int                                                                        | Integrations/Exp         | Total Integrations         | Exposures/Dith      | Total Dithers      | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                           | FASTR1             | 111                                                                               | 16                       | 32                         | 1                   | 2                  | 9940.193            |                              |

Proposal 9425 - Observation 2 - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

|                      |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| Special Requirements | Background Limited. Background no more than 10th percentile above minimum |
|----------------------|---------------------------------------------------------------------------|



# Proposal 9425 - Observation 3 - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

|                   |                                                                                                                                   |                    |                                                                                  |                          |                            |                     |                    |                     |                              |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------|--------------------------|----------------------------|---------------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 9425, Observation 3: Panoramic-z15<br>Diagnostic Status: Warning<br>Observing Template: MIRI Low Resolution Spectroscopy |                    |                                                                                  |                          |                            |                     |                    |                     | Mon Nov 10 23:00:30 GMT 2025 |
| Diagnostics       | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                       |                    |                                                                                  |                          |                            |                     |                    |                     |                              |
| Fixed Targets     | #                                                                                                                                 | Name               | Target Coordinates                                                               | Targ. Coord. Corrections |                            |                     | Miscellaneous      |                     |                              |
|                   | (3)                                                                                                                               | Panoramic-z15      | RA: 22 17 0.0874 (334.2503642d)<br>Dec: +00 22 45.13 (.37920d)<br>Equinox: J2000 |                          |                            |                     |                    |                     |                              |
|                   | Comments:<br>Category=Galaxy<br>Description=[High-redshift galaxies]                                                              |                    |                                                                                  |                          |                            |                     |                    |                     |                              |
| Acquisition       | #                                                                                                                                 | Target             | Filter                                                                           | Readout Pattern          | Groups/Int                 | Integrations/Exp    | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                                                                 | 4 TA_Panoramic-z15 | F560W                                                                            | FAST                     | 4                          | 1                   | 1                  | 11.1                | 251072                       |
| Template          | Subarray                                                                                                                          |                    |                                                                                  |                          | Obtain Verification Image? |                     |                    |                     |                              |
|                   | FULL                                                                                                                              |                    |                                                                                  |                          | false                      |                     |                    |                     |                              |
| Mosaic            | Rows                                                                                                                              | Columns            | Row Overlap %                                                                    | Column Overlap %         | Row shift (deg)            | Column shift (deg)  | Tile Order         |                     |                              |
|                   | 1                                                                                                                                 | 2                  | 0.0                                                                              | 80.0                     | 0.0                        | 0.0                 | DEFAULT            |                     |                              |
| Dithers           | #                                                                                                                                 | Dither Type        | No. Spectral Steps                                                               | Spectral Step Offset     | No. Spatial Steps          | Spatial Step Offset |                    |                     |                              |
|                   | 1                                                                                                                                 | ALONG SLIT NOD     |                                                                                  |                          |                            |                     |                    |                     |                              |
| Spectral Elements | #                                                                                                                                 | Readout Pattern    | Groups/Int                                                                       | Integrations/Exp         | Total Integrations         | Exposures/Dith      | Total Dithers      | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                                                                 | FASTR1             | 112                                                                              | 12                       | 24                         | 1                   | 2                  | 7520.358            |                              |

Proposal 9425 - Observation 3 - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

|                      |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| Special Requirements | Background Limited. Background no more than 30th percentile above minimum |
|----------------------|---------------------------------------------------------------------------|

# Proposal 9425 - Observation 4 - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

|                   |                                                                                                                                   |                    |                                                                                                                                                              |                            |                    |                     |                    |                     |                              |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------|---------------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 9425, Observation 4: Panoramic-z15<br>Diagnostic Status: Warning<br>Observing Template: MIRI Low Resolution Spectroscopy |                    |                                                                                                                                                              |                            |                    |                     |                    |                     | Mon Nov 10 23:00:30 GMT 2025 |
| Diagnostics       | (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                       |                    |                                                                                                                                                              |                            |                    |                     |                    |                     |                              |
| Fixed Targets     | #                                                                                                                                 | Name               | Target Coordinates                                                                                                                                           | Targ. Coord. Corrections   |                    |                     | Miscellaneous      |                     |                              |
|                   | (3)                                                                                                                               | Panoramic-z15      | RA: 22 17 0.0874 (334.2503642d)<br>Dec: +00 22 45.13 (.37920d)<br>Equinox: J2000<br><br>Comments:<br>Category=Galaxy<br>Description=[High-redshift galaxies] |                            |                    |                     |                    |                     |                              |
| Acquisition       | #                                                                                                                                 | Target             | Filter                                                                                                                                                       | Readout Pattern            | Groups/Int         | Integrations/Exp    | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                                                                 | 4 TA_Panoramic-z15 | F560W                                                                                                                                                        | FAST                       | 4                  | 1                   | 1                  | 11.1                | 251072                       |
| Template          | Subarray                                                                                                                          |                    |                                                                                                                                                              | Obtain Verification Image? |                    |                     |                    |                     |                              |
|                   | FULL                                                                                                                              |                    |                                                                                                                                                              | false                      |                    |                     |                    |                     |                              |
| Mosaic            | Rows                                                                                                                              | Columns            | Row Overlap %                                                                                                                                                | Column Overlap %           | Row shift (deg)    | Column shift (deg)  | Tile Order         |                     |                              |
|                   | 1                                                                                                                                 | 2                  | 0.0                                                                                                                                                          | 80.0                       | 0.0                | 0.0                 | DEFAULT            |                     |                              |
| Dithers           | #                                                                                                                                 | Dither Type        | No. Spectral Steps                                                                                                                                           | Spectral Step Offset       | No. Spatial Steps  | Spatial Step Offset |                    |                     |                              |
|                   | 1                                                                                                                                 | ALONG SLIT NOD     |                                                                                                                                                              |                            |                    |                     |                    |                     |                              |
| Spectral Elements | #                                                                                                                                 | Readout Pattern    | Groups/Int                                                                                                                                                   | Integrations/Exp           | Total Integrations | Exposures/Dith      | Total Dithers      | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                                                                 | FASTR1             | 112                                                                                                                                                          | 12                         | 24                 | 1                   | 2                  | 7520.358            |                              |

Proposal 9425 - Observation 4 - PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn

|                      |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| Special Requirements | Background Limited. Background no more than 30th percentile above minimum |
|----------------------|---------------------------------------------------------------------------|